Here is your presentation script and tactical execution checklist for the **6-minute final demo.** To land a perfect evaluation score, your delivery must focus heavily on **resilience, edge cases, and safe rejection**.

⏱️ The 6-Minute Script & Live Execution Flow

text

```
⏱️ MINUTE 0:00 – THE CORE MISSION (The Opening Argument)
"Good afternoon. Today, I am demonstrating FlyRank's AI Image Understanding & Content Matc
The core engineering philosophy here is that a production AI system shouldn't just be an u
It must treat safe rejection as a first-class feature. Let's look at how the system behave
```

Use code with caution.

text

```
⏱️ MINUTE 1:00 – PROBE 1 & 6: BATCH PROCESSING & TRACKING
[Action: Run `python -m engine.seed` in your terminal. Show logs streaming synchronously.]
"We are processing our corpus of roughly 50 animal images. As you can see, every image goe
asynchronous background pipeline with exponential backoff retries. The Gemini 2.5 Flash mo
rigidly validated JSON schemas that are parsed directly into our local SQLite instance.

Notice asset 'img_23'—it was a blurry background photo. The model returned a confidence ra
of guessing, our Pydantic validation layer immediately flagged it, isolated it, and threw
Simultaneously, we stream cost metrics per call—capturing exact input/output tokens and tr
```

Use code with caution.

text

```
⏱️ MINUTE 2:30 – PROBE 2: THE HAPPY PATH & CONCEPT MATCHING
[Action: Execute a GET request to `/posts/1/images` with text describing "the habitats of
"Next, let's examine our semantic search stream. Our blog post contains the scientific ter
Our local sentence-transformers model vectorizes this text in real time and multiplies it

The output is deterministic: our red fox asset surfaces in first place with a cosine simil
```

```
Even though the article did not use the phrase 'red fox', the engine matched the underlyin
Notice that our wolf and dog assets rank significantly lower down the vector chain."
```

Use code with caution.

text

```
⏱ MINUTE 3:45 – PROBE 3: THE FORCED-WOLF MOMENT (The Demo Highlight)
[Action: Manually update the post parameters to evaluate the grey wolf asset directly agai
"Now, let's push the engine into its most critical edge case: forcing a close canine looka

Watch the screen: the engine immediately issues a flat refusal. The Mismatch Guard interce
before a human operator ever sees it. The system provides a precise, clear explanation on
'REJECTED. Category Mismatch: Expected fox family variant, rejected grey wolf concept prof
This is the explicit separation between an unstable AI prototype and a reliable, productio
```

Use code with caution.

text

```
⏱ MINUTE 4:45 – PROBE 4 & 5: CONFIDENT ABSENCE & THE REVIEW API
[Action: Query an article discussing 'Deep sea oil rigs' against your animal image library
"What happens when the system encounters a completely out-of-domain article? Let's check i
The system does not force an incorrect recommendation. It safely returns: 'No confident ma
Highest similarity score of 0.28 failed to clear the precision gate boundary of 0.54.'

We can then toggle over to our Review API endpoint. If an operator manually approves or re
our persistent relational schema records the audit trail, maintaining data consistency acr
```

Use code with caution.

text

```
⏱ MINUTE 5:45 – THE CLOSING NUMBERS (Your Final Line)
"To defend this system, I built an automated precision sweep script to map our decision bo
Across our expanded evaluation dataset, this exact combination of structural schemas and v
yielded a Top-1 Precision Score of exactly 92% [or insert your exact script number here].
```

```
Good suggestions when confident, safe rejection when uncertain—that is production AI. Than
```

Use code with caution.

📋 Pre-Flight Verification Checklist

Before joining the call with your FlyRank technical evaluator, run through these diagnostic steps to ensure everything passes smoothly:

- **Local DB Reset:** Clear your local `.db` file using a quick script to guarantee your seed metrics start completely fresh.
- **The Zero-Token Lockout:** Confirm your `GEMINI_API_KEY` environmental parameter is loaded correctly inside your shell.
- **EVIDENCE.md Match:** Verify that the precision numbers displayed in your terminal match the values written in your `README.md`.
- **The Code Deep-Dive:** Be prepared to show your exact `evaluate_mismatch_guard` code block. The evaluator will ask you to explain its logic on the spot to verify code ownership.

To help you put the final touches on your code before the demo, what should we build next?

↪ Build the automated **mock database seeder script** (`engine/seed.py`) to satisfy the `capstone.yaml` execution command

↪ Build a **PyTest script** to verify that your endpoint boundaries throw a 400 error on empty text inputs